SYSTEM DESIGN: ZERO TO INTERVIEW HERO

A–Z System Design Glossary (Whole Course)

Every term from all 20 chapters in plain words, in one searchable place. This isn't just for the lecture you're on, so keep it handy for the whole course.

By Nata Vacheishvili · udemy.com/course/system-design-zero-to-hero

HOW TO USE IT

Look anything up

- Press Ctrl+F (Cmd+F on a Mac) to search for a term.
- "Taught in 4.2" means the second lecture of Chapter 4. Some terms appear more than once, because a later chapter adds to them or uses them differently.
- 705 terms from all 20 chapters.

#

304	A status code that means not modified. `Taught in 9.6`
404	A status code that means not found. `Taught in 8.7`
429	The HTTP code for too many requests. `Taught in 4.3`

A

A*	A search that picks the smallest time so far plus a guess of the time that's left. Say it as A star. `Taught in 18.7`
Acceleration limits	Limits on a sharp increase in usage. Our reasoning is that chips can't just appear in a minute. `Taught in 20.7`
Access control list (ACL)	A list of who can do what. `Taught in 15.8`
ACID / transaction	Atomic, consistent, isolated and durable. A group of steps that works like this is called a transaction. `Taught in 15.3`
Acknowledgment	The "got it" message a server sends back. `Taught in 6.4`
Acks	Short for acknowledgments. Kafka's setting for how long a producer waits for got it. All is the default today. `Taught in 19.10`
Adaptive bitrate streaming	The quality adapts to your connection, one small request at a time. `Taught in 14.3`
Adjacency list	An address book for each junction, listing its roads going out. `Taught in 18.7`
AES	The Advanced Encryption Standard, published by the United States government in 2001, a strong and well-tested way of scrambling data. `Taught in 14.8`
Agent	A tool that acts for a person. `Taught in 9.9`
Aggregate location data	Data from a lot of phones added together. `Taught in 18.9`
Aggregated and anonymized	Only totals, with no names on it. `Taught in 16.11`
Aggregating	A fancier word for adding things up. `Taught in 13.7`

AirTag	Apple's tracker tag, about the size of a coin. `Taught in 17.10`
AJAX	An old name for a page asking for something in the background, without reloading. `Taught in 13.8`
Algorithm	A step by step way of doing something. `Taught in 4.2`
Allocated	How much space we've promised, which isn't how much is being used. `Taught in 15.1`
Ambiguity	The question is vague, and it could mean a lot of different things. `Taught in 3.1`
Analytics	Counting and studying what people do. `Taught in 8.2`
Ancestors	The dots on the path above you, like your parent and your parent's parent. `Taught in 13.6`
Anti-entropy	Servers quietly comparing and fixing their copies in the background. Entropy is things slowly drifting into disorder. `Taught in 6.9`
Apache Cassandra	A popular open source database whose machines sit on a ring. It calls each spot a token. `Taught in 5.5`
API	A way for one program to request something from another. `Taught in 1.1`
API gateway	A service in front of your APIs that handles jobs they all need, like checking who's calling, and rate limiting. `Taught in 4.3`
APNs	Apple Push Notification service. The service that delivers messages to iPhones. `Taught in 10.2`
Appeal	Someone can ask for another look. `Taught in 14.9`
Approximate location	A blurry location. Android's docs say it's within about 3 square km, and Apple has said a typical value is around 5 km. `Taught in 16.11`
As the crow flies	Straight line distance. `Taught in 16.8`
ASCII	An old table where every character is a number, so capital A is 65. It only has 128 characters, so each fits in 1 byte. `Taught in 13.10`
Assumption	A number you decide to go with because you don't know the real one. You share it with your interviewer. `Taught in 2.1`
Asynchronous	Sender and receiver don't have to work at the same time. Like texting a friend. `Taught in 1.9`
At least once	We keep trying until we know a message was accepted, so it might arrive more than once. `Taught in 10.6`
At most once	Each message gets processed 1 time or 0 times. `Taught in 19.10`
Atomic	It either happens all at once, or it doesn't happen at all, and nobody can see it halfway through. `Taught in 4.7`
Auth token	A long secret string our server gives an app at login, which basically says, this is Ben. `Taught in 11.2`
Authentication	Proving who you are. `Taught in 10.2`

Auto increment	The database has a counter. It gives a new row the next number, then adds 1. Taught in 7.1
Auto scaling	Adding or removing web servers automatically based on traffic. Taught in 1.8
Autocomplete	The little list under a search box. You'll also hear typeahead, and search as you type. Taught in 13.1
AV1	A newer codec from the Alliance for Open Media, free to use. Taught in 14.2
Availability	How much a system is up, written as a percentage of uptime. Taught in 2.4
Availability (in CAP)	Every request gets an answer and not an error, even if the answer is out of date. Taught in 6.2
AWS	Amazon Web Services. A service provider that offers infrastructure like servers, storage and databases. Taught in 2.4

B

B-tree	The common kind of tree in databases, with a lot of branches on each level. It's the default in PostgreSQL, and you most probably won't be asked how it works. Taught in 13.3
Back of the envelope calculation	A quick, approximate estimate, with math simple enough to do on the back of an envelope. The goal is a feel for the scale, not precision. Taught in 2.1
Back queues	The per-website lines: being polite. Taught in 9.6
Backend	Everything that runs on our side, like the servers and the data. Taught in 4.1
Background	You're not looking at the app, so it's in your pocket. Taught in 17.8
Bandwidth	How much data can move through a connection each second. Taught in 2.2
Base	How many digits you have before a wheel rolls over. Base 10 has 10, and base 2 has 2. Taught in 8.5
Base 32	32 symbols on each wheel of our odometer. Each character is 5 switches. Taught in 16.5
Base 62	Counting with 62 digits: 0 to 9, a to z, then capital A to Z. Taught in 8.5
Batch	A pile of work that you do all at once. Taught in 13.7
Batch API	You send a huge pile of requests, and you get the answers back within a day, for half the price. Taught in 20.9
Bidirectional	Two-way, so both sides can send. Taught in 12.2
Big O	A short way of saying how much the work grows when the data gets bigger. Taught in 13.2
Binary	Zeros and ones, which is how computers operate. Taught in 2.2
Binary file	A compact format made for computers, and not text a person would read. Taught in 18.5

Birthday surprise	With just 23 people in a room, it's about a coin flip that 2 share a birthday. Collisions come much sooner than you'd expect. Taught in 8.4
Bit	A single zero or one. Taught in 2.2
Bitrate	How much data a video uses for each second. Like water going through a pipe. Taught in 14.2
Black box	Something we use from the outside, and we don't care what's inside. Taught in 15.2
Blob storage	Another name for object storage, used by Microsoft's cloud. A blob is just a big file. Taught in 14.4
Block	A piece of a file. Each piece is the same size, and each one gets a fingerprint. Taught in 15.4
Block server	One of our own servers that sits in the middle and takes the blocks. Taught in 15.5
Bloom filter	A row of lights in memory that answers definitely not, or maybe. Named after Burton Bloom. Taught in 6.11
Blue green	You build a whole new set right beside the old one, check it, and then switch people over. Taught in 16.8
Bluetooth	The short-range radio that connects your headphones to your phone. Taught in 17.10
Bookmark	Our name for one small number a device keeps for a chat: the last message number it has. Taught in 12.6 The number that each reader keeps, to know where it's up to. Taught in 19.2
Bot	A program that visits websites by itself. A crawler is one. Taught in 9.5
Bottleneck	The one part of a system that slows everything else down, like one narrow door in a very busy store. Taught in 3.5
Breadth first	Read everything on this page first, then everything those pages link to. Breadth means how wide something is. Taught in 9.4
Broker	A go-between that takes messages from one side and hands them to the other. Taught in 19.1
Brute force	Trying every possible answer until one of them works. Taught in 4.1
Buffer	A small area in memory, like the 2 trays on the desk. In Mercator, those trays held only 600 links each. Taught in 9.7 The little stock of video that the player keeps ready to play. Taught in 14.3
Burst	A lot of requests arriving at once. Taught in 4.5
Byte	8 bits. Enough for about one English letter. Emojis and many other alphabets take a bit more. Taught in 2.2

C

Cache	Temporary storage for frequently used data, or responses that were expensive to get. Taught in 1.6
Cache miss	The cache doesn't have the answer, so we have to go to the database. Taught in 5.1
Cache-Control	A header that controls how long a browser remembers an answer. Taught in 8.2
Candidates	The posts that could possibly be shown. Taught in 11.7 A few good routes that the route planner finds. You might hear this called top K shortest paths, where K is just a number. Taught in 18.10
CAP theorem	When servers can't talk to each other, you have to choose between staying correct and staying available. Taught in 6.2
Capacity	How many places a square can hold. It's a number we pick when we build the tree. Taught in 16.7
Capacity estimate	Figuring out how much storage or memory a system needs. Taught in 2.2
CDN	Content delivery network. Servers spread across locations that deliver static content. Taught in 1.7
Cell tower	The antenna that your phone connects to for calls and data. Taught in 17.1
Certificate	Like an ID card for our server. Taught in 10.2
Change list	A list for each shared folder, where every commit adds one line with the next number. Taught in 15.6
Channel (pub-sub)	The station in pub-sub. Anyone tuned in hears what's sent, and it doesn't keep anything. Taught in 17.4
Checkpoint	Writing down where you are on storage that survives the machine. It's like saving your game. Taught in 9.7
Children / parent	The dots right below a dot are its children, and the dot right above is its parent. Taught in 13.5
Chip	The piece of hardware that does the actual computing. Taught in 14.6
Chromium	The free, open source browser project that Google's Chrome browser is built on. Taught in 9.8
Client	The computer or app asking the server for something. Taught in 1.1
Clockwise	The direction that the hands of a clock move. Taught in 5.2
Cloudflare	A company that a huge number of websites put in front of their servers, to protect them from attacks and speed them up. Taught in 9.9
CloudFront	Amazon's CDN. Its public price page for the United States starts at about 8.5 cents a gigabyte, and goes down to 2 cents past 5 petabytes in a month. Taught in 14.1
Cluster	A big group of machines being run together. Taught in 20.10

Clustering key	The part of a key that sorts things inside a folder. The partition key decides the folder. Taught in 18.9
CMAF	A format that lets both menus point at one set of files. You probably won't get asked about it. Taught in 14.3
Codec	The method for squeezing a video, from coder and decoder. Some detail is thrown away for good. Taught in 14.2
Coins	Our name for a rate limiter's tokens, which have nothing to do with text. I'd say that out loud in your interview. Taught in 20.7
Cold storage	Cheaper storage that's slower to get back. Cold just means rarely touched. Taught in 15.9
Collapse ID	If you send a notification with one, Apple can merge it with an earlier one that has the same ID. Google has something similar. It won't help with a text. Taught in 10.6
Collision	2 different things get the same ID. Taught in 7.1
Commit	The last step of the conversation, which means that it's now real. Taught in 15.5
Commit log	The notebook on disk. Every change gets added at the very end, so it survives a power cut. Taught in 6.10
Committed (a message)	When everyone on the in-sync list has it, so it's safe. Taught in 19.9
Committing (a bookmark)	A reader saving its place by telling the coordinator, I've finished up to here. Taught in 19.8
Common Crawl	A non-profit that crawls the web and gives the results away for free, so researchers don't need a crawler of their own. Taught in 9.1
Compaction	Merging a few boxes into one new box and throwing the old ones away. Taught in 6.11 Asking the model to write a short summary of the old part, and sending that instead. Taught in 20.5
Compare-and-set	One step that nothing can interrupt: set the latest to 10, only if it's still 9. One wins, and the other gets told no. Taught in 15.7
Compiler	The program that turns code into something a chip can run. Taught in 20.10
Composite key	A key made of two parts. Taught in 12.4
Compression	Squeezing data to take less space, like a zip file. Taught in 6.1
Compute power	The CPU and memory that do the actual work. Taught in 4.1
Conflict	Neither version is older. Each one knows about a write the other one doesn't. Taught in 6.6
Conflicted copy	Dropbox's name for the file that keeps the loser's work, right next to the original. Taught in 15.7
Connection server	Our name here for the servers that hold everybody's open lines, like WhatsApp's chat servers. Taught in 17.4
Consistency (in CAP)	Everyone sees the same, latest data. Taught in 6.2

Consistent hashing	Finding a server by walking around a ring. It's called consistent because when the servers change, most of the answers stay the same. Taught in 5.2
Consumer group	A group of readers that share the work, where each partition goes to only one member. Taught in 19.4
Consumer lag	How far a reader's bookmark is from the end of the book. It's probably the first thing I'd put on a dashboard. Taught in 19.4
Container	The box that holds the video, the sound and maybe subtitles. That's what .mp4 is, so it doesn't tell you the codec. Taught in 14.2
Content ID	YouTube's system for finding copies of other people's work. Copyright owners submit their files to a database, and new uploads are compared to it. Taught in 14.9
Content Seen	The box that asks: have we already seen these exact words somewhere else? Taught in 9.2
Content-defined chunking	Cutting a file by what's in the content, instead of by position. Taught in 15.4
Context	Everything that the model has in front of it. Taught in 20.2
Context window	The total a model can look at, including the answer it's writing. Taught in 20.5
Continuous batching	Letting people step on and off between steps. The paper called it iteration-level scheduling, where iteration means one trip around the loop. Taught in 20.6
Continuous integration	Each code check-in is automatically verified, so teams detect problems early. Taught in 1.9
Contraction hierarchies	A research idea where you add shortcut arrows ahead of time. You probably won't be asked how it works. Taught in 18.7
Control plane	The part of Kubernetes that makes those decisions, like a manager's office. Taught in 20.10
Controller	The machine with the job of deciding who's the new leader. Taught in 19.9
Coordinator	Whichever server takes the request, like a waiter passing your order to the kitchen. Any server can do this. Taught in 6.4
Corrupt	Damaged. A corrupt file fails on every worker. Taught in 14.6
CP / AP	CP stays correct during a partition. AP stays available. Taught in 6.2
CPU	Central processing unit. The brain of a computer, which runs instructions and calculations. Taught in 1.3
CPU bound	When the thing that limits you is how fast the machine can work, and not how much it can hold. Taught in 17.6
CPU cache	A tiny, very fast memory right next to the processor. Taught in 2.3
CRC	Cyclic redundancy check. A small number worked out from a batch's bytes and saved with them. If it doesn't match later, something got damaged. Taught in 19.6
Cron job	A program set to run on a schedule, like the alarm on your phone. Taught in 10.1
CRUD	Create, read, update and delete. Taught in 1.1

CSS	The code that controls how a website looks. `Taught in 1.7`
Cursor	A little marker that says where we stopped, so that new posts on top don't shift the page. `Taught in 11.6`

D

Daily active users	People who open the app on any given day. `Taught in 2.5`
Dark launch	Running a new limit so it decides but doesn't block anybody, and only writes down what it would have blocked. Also called shadow mode. `Taught in 4.8`
DASH	An international standard that no single company owns, and it became a standard in 2012. It does the same job as HLS, and its menu just looks different. `Taught in 14.3`
Data center	A building full of servers. `Taught in 1.8`
Data retention	How long we keep the data for. The longer the retention, the larger the total storage we need. `Taught in 2.6`
Data scraping	Using programs to automatically copy large amounts of data from a website. `Taught in 4.1`
Data structure	A way of arranging data so that one kind of question gets easy. `Taught in 13.3`
Database replication	Keeping copies of your database on more than one server. `Taught in 1.5`
Dead letter queue	A separate line for messages we gave up on, so they don't block anyone, and a person can look at them. `Taught in 10.5`
Dead reckoning	An old word from ships, for working out where you are from your speed and direction when you can't see anything. `Taught in 18.8`
Debouncing	Only sending a request once someone has stopped typing for a moment. Like an elevator door, which only closes when nobody's come in for a little while. `Taught in 13.8`
Decoupling	The sender doesn't care how fast the receiver is, or whether it's online. `Taught in 1.9`
Deduplication	Keeping one copy of something that shows up many times. Also called dedupe. `Taught in 15.5`
DeepMind	Google's AI research lab. `Taught in 18.10`
Delayed message	A message that should show up later, and not now. `Taught in 19.11`
Delta sync	You only send the difference. Delta is a word for the difference between 2 things. `Taught in 15.4`
Denial of service attack (DoS)	Flooding a system on purpose, so that real users can't get in. `Taught in 4.1`
Denormalization	Intentionally adding duplication, so queries don't have to join many tables. `Taught in 1.10`
Depth first	Keep going deeper before you ever come back up. `Taught in 9.4`

DESC	Short for descending, so biggest first. Taught in 13.4
Design deep dive	Going into a lot more detail on the parts that matter. Taught in 3.4
Detecting Unwanted Location Trackers	A shared specification from Apple and Google. If an unknown tag is seen moving with you, your phone can warn you. Taught in 17.10
Device token	A long string of letters and numbers that Apple or Google gives to our app on a phone. A delivery address only the courier understands. Taught in 10.2
Dijkstra's algorithm	A search that always finishes the closest unfinished junction next, named after Edsger Dijkstra, a Dutch computer scientist. Taught in 18.6
Directed acyclic graph (DAG)	Dots and arrows where the arrows go one way (directed), and you can't follow them back to where you started (acyclic). Taught in 14.5
Disk	Long-term storage, like hard drives or solid state drives. Data stays there even after the power is off, but it's much slower to access. Taught in 2.3
Disk seek	When a hard drive has to move to the right spot before it can start reading. Taught in 2.3
Distributed environment	Running on many machines instead of one. Taught in 4.2
Distributed message queue	A message queue that runs across a lot of machines. Taught in 19.1
Distributed storage	Spreading your data across many machines. Taught in 2.1
DNS	Domain Name System. Translates names like google.com into IP addresses, like the contacts on your phone. Taught in 1.1
Downloader	The program that actually fetches pages. Taught in 9.2
Downstream	Anything further along that the request would have gone to next. Taught in 4.3
Draining	Telling a server to stop taking new people, and waiting for it to empty. Taught in 12.3
DRM	Digital rights management. How paid films usually work. Taught in 14.8
Durability	Once it's saved, we don't lose it. Taught in 15.1
Durable	Messages are safely kept, so they aren't lost. Taught in 1.9
Dynamic rendering	A website sends crawlers a different version of its page. Taught in 9.8
Dynamo	A storage system Amazon built for its own use, including its shopping cart. It uses a ring with virtual nodes. Taught in 5.5

E

E.164	The standard phone number format: a plus, a country code, then the rest of the number, with at most 15 digits. Taught in 10.3
EC2	The AWS service that rents out servers. Taught in 2.4

Edge location	One of many locations close to users where a system runs something, like limiters. `Taught in 4.7`
Elasticsearch	A popular search engine you can add to your own system, with a built-in feature for search as you type. `Taught in 13.4`
Elixir	A newer language that runs on the same engine as Erlang. `Taught in 17.7`
Encoding	Squeezing a video with a codec. `Taught in 14.2`
Encoding on demand	Making a version only when someone asks for it and it isn't there. `Taught in 14.7`
Encrypted	Scrambled, so that it can't be read without a key. `Taught in 2.6`
Encrypted at rest and in transit	Encrypted on the company's disks and on the road. `Taught in 12.9`
Encryption at rest	Each block gets scrambled before it's stored. The company holds the key, so it can still make previews, search and dedupe. `Taught in 15.8`
Encryption in transit	Encrypted on the road. That's HTTPS, and it stops anyone on the café wifi. `Taught in 15.8`
Encryption on the device	The file gets locked before it leaves Anna's phone, and the company never gets the key. `Taught in 15.8`
End-to-end encryption	A message stays locked the whole way from one device to the other, so nobody in between can get to the content. `Taught in 12.9`
Endpoint	One specific address you can send a request to. One item on the API's menu. `Taught in 8.2`
Epoch	The moment we start counting time from. It's like a birthday. `Taught in 7.5`
Equator	The ring around the very middle of the Earth. `Taught in 16.3`
Erlang	A programming language made for running a huge number of small things at the same time. `Taught in 17.7`
ETA	Estimated time of arrival. `Taught in 18.10`
etcd	A small and reliable key-value store for keeping that same kind of information, which also uses Raft. `Taught in 19.11`
Event ID	One ID for this one thing that happened, so that every retry carries the same ID. `Taught in 10.6`
Eventual consistency	Copies can disagree for a moment, but they catch up. `Taught in 4.7` If people stop making changes, all of the copies end up the same, with no exact time on it. `Taught in 6.5`
Eviction	Removing an existing item to make room for a new one. `Taught in 1.6`
EXPIRE	A Redis command that deletes a key after a certain number of seconds. `Taught in 4.6`
Expiry	A time after which a message isn't worth sending. `Taught in 10.7`
Explore and Reels	Instagram's discovery and short video sections. Mosseri says most of what you see there comes from accounts you don't follow. `Taught in 11.7`

Exponential backoff	Waiting longer after each failure: 1 second, then 2, then 4. Taught in 4.8
Extensibility	Designing so that you can add something new without redoing everything. Taught in 9.8

F

Failing closed	If the limiter breaks, reject every request. Taught in 4.8
Failing open	If the limiter breaks, let every request through. Taught in 4.8
Failover	If one machine fails, another takes over. Swapping in the spare tire. Taught in 1.3
False positive	The answer looked like a yes, and it was wrong. Taught in 6.11
Fan-out	One request or one thing spreading out to a lot of places, like a fan opening up. Taught in 11.3
Fan-out on read	Doing that spreading out when somebody reads. Taught in 11.3
Fan-out on write	Doing the spreading out when somebody writes a post. Taught in 11.4
FCM	Firebase Cloud Messaging. Firebase is Google's set of tools for people who build apps, and this is the part that delivers messages. Taught in 10.2
Feature flag	A switch that lets you turn new code on slowly. Taught in 20.10
Feed	The list you scroll when you open the app. It's different for every person, because it's built from the people they follow. Taught in 11.1
FFmpeg	A free tool for pretty much any job with video and sound files. Taught in 14.2
FIFO	First in, first out. Things leave in the order they arrived, like a line at a coffee shop. Taught in 4.5 First in, first out. A normal line, where whoever arrived first gets served first. Taught in 9.4
Filter	A short list of things we never show. Every answer goes past it on the way out. Taught in 13.9
Find Hub	Google's version of this network. Taught in 17.10
Find My	Apple's app for finding your devices, your things and your friends. Taught in 17.10
Fingerprint	A hash used as one short number that stands for something bigger. Change even one letter, and you almost certainly get a different number. Taught in 6.9
Firewall	Like a security guard at the edge of a network, who only lets certain traffic through. Taught in 12.2
Fixed window counter	Cuts time into windows, gives each customer a counter, and resets it when the next window starts. Taught in 4.4
Flink	Keeps running counts on a stream as it goes by. Taught in 13.10
Flush	Writing the desk out to disk. Taught in 6.10

For You feed	What you land on in TikTok: a stream of videos curated to your interests. `Taught in 11.7`
Foreground service	What an app runs to get more location updates. It shows a notification on the screen the whole time, so you know it's happening. `Taught in 17.8`
Foreign key	A column that points to a row in another table. `Taught in 10.3`
Forged	Faked. `Taught in 4.3`
Forward	One update being passed along to one friend's connection. `Taught in 17.2`
Frame	One still picture in a video, like one page of a flipbook. `Taught in 14.2`
Frame rate	How fast you flip, in frames per second. 30 or 60 is normal. `Taught in 14.2`
Frequency cap	A limit on how many messages of a certain kind one person gets in a day or a week. `Taught in 10.7`
Front queues	The importance lines: what matters. `Taught in 9.6`

G

Garbage collection	Sweeping up what nobody's using anymore, like the blocks from when Dan gave up in the tunnel. `Taught in 15.9`
GDPR	The European Union's privacy law, which names location data in its definition of personal data. Nothing here is legal advice. `Taught in 16.11`
Geocoding	Turning words into a latitude and a longitude. `Taught in 18.2`
GeoDNS	A DNS service that gives back a different IP address based on the user's location. `Taught in 1.8`
Geofence	A shape on a map with a rule attached, so that something happens when you go in or come out. `Taught in 16.9`
Geohash	An address for a square. Each cut writes down which piece you went into. `Taught in 16.4`
GEOSEARCH	A Redis command. Places are kept in a sorted set, a list in order by a score, with a geohash as that score. `Taught in 16.9`
GET / POST	GET means give me something. POST means here's something new, please save it. `Taught in 8.2`
Ghost Mode	Snapchat's setting that hides you from everyone on its map, with a timer of 3 hours, 24 hours, or until you turn it off. `Taught in 17.11`
Gigabyte (GB)	1,024 megabytes. Round it to a billion bytes. `Taught in 2.2`
GIS	Geographic information system. The general name for software that works with maps. `Taught in 18.2`
Glacier	Amazon's cold storage. The slowest one, Deep Archive, can take 12 hours to get something back, or up to 48 on its cheapest setting. `Taught in 15.9`

Google Drive / Dropbox	Google's service for keeping your files online, and a company built around the same idea. `Taught in 15.1`
Google Play services	Google's own software that comes on most Android phones. `Taught in 10.2`
Google-Extended	A setting in your robots file for whether what Google crawled can train Gemini, its AI model. `Taught in 9.9`
Gossip	Servers telling a few random servers what they know, so news spreads like an office rumor. That's the real technical word. `Taught in 6.7`
GPS	Global Positioning System. A group of satellites run by the US government, each one broadcasting where it is and what time it is. `Taught in 17.1`
GPU	Graphics processing unit. A chip that does a huge number of small calculations at once, first built to draw video game graphics. `Taught in 20.2`
Graph	Dots and arrows. It isn't the kind with a line going up from math class. `Taught in 9.4`
Graph database	A database built around dots and arrows. `Taught in 11.3`
Graph neural network	A kind of machine learning Google used with DeepMind. It fits because roads are a graph, and traffic on one road spills onto the next. `Taught in 18.10`
Group coordinator	One broker that looks after one consumer group. It knows who's in the group and where their bookmarks are. `Taught in 19.8`
Group leader	Whichever member got picked to make the plan. It has no other power. `Taught in 19.8`
Group of pictures (GOP)	A keyframe and the changes that follow it. `Taught in 14.2`

H

H.264	The older codec that almost everything can play. `Taught in 14.2`
H3	Uber's grid of hexagons covering the whole planet. `Taught in 16.9`
Hadoop and Spark	Open source tools for running big jobs across a lot of machines, so you don't have to write all that splitting up yourself. `Taught in 13.10`
Hallucination	When a model writes something that sounds completely sure of itself and is just wrong. `Taught in 20.2`
Handshake	A quick hello between two sides at the start of TCP. After that, they have a connection. `Taught in 12.2`
Hard limit	A limit that can never be passed. `Taught in 4.8`
Hash function	A formula that turns a value into a number. `Taught in 1.10`
Hash ring	The circle that servers and keys sit on. Also just called a ring. `Taught in 5.2`
Hash space	All of the possible spots together. Ours has 12, and real rings have billions and often way more. `Taught in 5.4`

Hash table	The dictionary programmers use all the time. A hash function gives you the hook number, so you jump straight there. Taught in 6.1
HBase	An open source database for spreading data over many machines. Taught in 12.4
Headers	Small pieces of information that travel along with a response. The user doesn't see them, but the program on the other side does. Taught in 4.6
Headless browser	A real browser without a screen. Taught in 9.8
Health check	A small request sent to each server on a schedule, to see if it's still answering. Taught in 1.4
Heap	A sorted list that gives you the earliest one quickly. Taught in 9.5 It only promises that one specific thing is always on top. Taught in 13.3
Heartbeat	A phone keeps sending a tiny message that says, still here. As long as it keeps going, you're alive. Taught in 12.8
Heartbeat counter	One number a server keeps about itself and adds to regularly. As long as it keeps going, you're alive. Taught in 6.7
Heatmap	A map of where people have been active, with busy places glowing brighter. Taught in 16.11
Helper	A small piece of code for each open line. It holds the line, remembers that person's newest spot, and subscribes to their friends' channels. Also called a connection handler. Taught in 17.4
Herd effect	Google's name for programs all coming back at the same time and hitting the same thing. It's our retry storm. Taught in 20.10
HEVC	A newer codec, also called H.265, with licensing fees. Taught in 14.2
Hexadecimal	Counting with 16 symbols, 0 to 9 and then a to f. Each symbol stands for 4 bits. Taught in 7.3
Hierarchy	Levels. Here, small roads, bigger roads and highways. Taught in 18.7
High availability	A system stays up and running without interruption for as long as possible. Taught in 2.4
High level	The big picture: the main boxes and how they connect, without the details of each one yet. Taught in 3.3
High watermark	The line that readers can read up to. You only need to recognize the name. Taught in 19.9
Hilbert curve	A line you can draw without lifting your pen. It goes through every square once, and every step goes to a square that touches the last one. Taught in 16.9
Hint	A note saying this copy really belongs to another server. Taught in 6.8
Hinted handoff	A server holds a copy with a hint, then hands it back when the real owner returns. Taught in 6.8
HLS	HTTP Live Streaming, which came from Apple. A menu and pieces. Taught in 14.3

Hot partition	Everyone is reading the same busy chunk of data, so it all lands on one partition. It's one very popular key again. `Taught in 12.4`
Hotspot key	Another name for the celebrity problem. `Taught in 1.10`
HTML	The file that describes what a page looks like. `Taught in 1.1`
HTTP request	A message your computer sends to a web server asking for something. `Taught in 1.1`
HTTPS	The secure kind of web connection. `Taught in 11.2`
Hybrid	Most people's posts get written ahead of time, and celebrity posts get fetched when you open the app. `Taught in 11.5`
Hydration	A funny word for filling in the rest. You start with something dry, only the numbers, and you add everything else. `Taught in 11.6`

I

I/O	Input and output. Reading from and writing to something like a disk. `Taught in 19.5`
Idempotency	Doing something twice has the same effect as doing it once. `Taught in 7.3`
Idempotent producer	If a producer sends again, the broker notices the repeat, and only 1 copy goes in the log. It's on by default now. `Taught in 19.10`
Immutable	Can't be changed. `Taught in 6.10`
In-sync replicas (ISR)	A live list the leader keeps of who's caught up. `Taught in 19.9`
INCR	A Redis command that adds 1 to a counter and gives you back the new number. If the counter doesn't exist yet, it starts from 0. `Taught in 4.6`
Index	A sorted list a database keeps so it can find a row without reading the whole table, like the list of topics at the back of a book. `Taught in 7.3`
Inference	Using the model to answer someone. It just means running the loop. `Taught in 20.2`
Input / output	The text you send in, and what the model writes. `Taught in 20.2`
Input method editor	You type on a normal keyboard with letters, then choose the characters you meant. Often used in languages like Chinese and Japanese. `Taught in 13.10`
Internal node	A dot with children. It only holds directions. `Taught in 16.7`
Internet Archive	A non-profit library that saves copies of web pages. Its tool for looking at old copies is called the Wayback Machine. `Taught in 9.1`
Interpolation	Working out a spot between 2 you know, like number 50 being about halfway along a block. `Taught in 18.2`
Invalidate	Tell the CDN to remove a file. `Taught in 1.7`
IP address	A set of numbers that tells other machines where to find a machine, like a home address. `Taught in 1.1`

ISP	Internet service provider. A company that sells people their internet connection. Taught in 14.7
Item	The things you see in a folder. A file and a folder are both items. Taught in 15.3

J

Jitter	A little randomness added to a wait, so that retries don't all come back together. Taught in 10.5
Join	Combines rows from two or more tables based on a related column. Taught in 1.2
JSON	JavaScript Object Notation, a simple text format for sending data. Taught in 1.1
Junction	A place where roads meet. Taught in 18.6

K

K nearest	Finding the closest K of something, however far away. K is just a number. Taught in 16.8
Kafka	Carries a huge stream of events from the programs that write them to the programs that read them, kind of like a giant shared diary. Taught in 13.10
Keep-alive	Normal HTTP connections staying open. The server still can't say anything until it's asked. Taught in 12.2
Key	Anything we look up, like Anna. Taught in 5.4
Key-value store	Saves values under unique keys, like a dictionary. It only knows how to find things by their key. Taught in 6.1
Keyframe	A whole picture. Everything after it is just a list of changes. Taught in 14.2
Kilobyte (KB)	1,024 bytes. Round it to a thousand bytes. Taught in 2.2
KRaft	The design where Kafka does ZooKeeper's job itself, using Raft. Taught in 19.11
Kubernetes	A very popular open source system for running programs across many machines. It starts them, restarts them when they crash, and moves them around. Taught in 20.10
KV cache	The model's notes. It stands for key-value cache, and it has nothing to do with our key-value store chapter. Taught in 20.2

L

Language of nines	Describing uptime by counting the nines. 99.9% is three nines. Taught in 2.4
Last write wins	Look at the time and keep the newest version. It's simple, but it can lose an update. Taught in 6.5
Latency	The delay between sending a request and receiving the response. Taught in 1.2

Latitude	How far north or south you are. The equator is zero, and the poles are 90 degrees north and 90 degrees south. Taught in 16.3
Layer 3	The network layer, where we can limit by IP address. Taught in 4.8
Layer 7	The application layer, where we look at HTTP requests. Taught in 4.8
LBFS	A research file system built for slow networks, which described this idea in 2001. You probably won't get asked this. Taught in 15.4
Leaky bucket	A bucket that drains at a steady speed and overflows if you pour too fast. Either a real queue, or just a meter. Taught in 4.5
Leaves	The bottom of the tree, which are the shelves here. Taught in 6.9
Library	Code that someone wrote so that you don't have to. Taught in 16.9 Ready-made code that you add to your program, so you don't have to write it. Taught in 19.7
License server	Checks that you're allowed to watch, and sends back the key and the rules. Taught in 14.8
LIKE	In a database query, it means match a pattern. The percent sign means anything at all can come after this. Taught in 13.4
List (array)	Like a row of numbered shelves, sitting side by side. Taught in 13.3
Load balancer	Takes every incoming request and spreads the requests across servers. Like the host at a busy restaurant. Taught in 1.4
Location header	A header that holds the address to go to. Taught in 8.2
Location-based service (LBS)	A name for anything that answers based on where you are. Taught in 16.2
Lock	One server says, nobody touch this until I'm done. Taught in 4.7
Log	A record of what happened. Taught in 1.9 A record of what happened. Here it has 2 rules: only add to the end, and never change what's written. Taught in 19.2
Logarithm	How many times can you cut this in half until 1 is left? Also shortened to log. Taught in 13.2
Long polling	You ask, and the server holds on to your request until something happens or time runs out. Taught in 12.2
Long tail	The thin part of the chart, with the very long list of things that hardly anyone wants. Taught in 14.7
Longitude	How far east or west you are, up to 180 degrees east and 180 degrees west, which meet on the other side of the world. Taught in 16.3
Low-Latency HLS	An extension of HLS where the player can get small parts of a piece before the piece has finished. You probably won't get asked how it works. Taught in 14.9
LRU	Least recently used. Removes whatever hasn't been used for the longest time. Taught in 1.6

Lua script	A small program you can run in Redis, as one atomic step. Taught in 4.7

M

Machine learning model	A program that learned its rules from a huge number of examples, instead of someone writing them by hand. Taught in 11.7
Magic Pocket	Dropbox's own custom-built storage. Taught in 15.9
Mail Privacy Protection	Apple's feature to stop senders from learning when you read things. It downloads those images in the background, whether or not you open the email. Taught in 10.8
Majority	More than half. Only one side of a break can ever have one. Taught in 6.2
Map matching	Taking a dot and putting it on a road, like a spell check with a dictionary of all the roads. Also called snapping to a road. Taught in 18.8
Map tile	One of the small squares the map is cut into, like a tile in a mosaic. Taught in 18.4
Master / slave	The original, which takes writes, and the copies, which serve reads. Also called primary and replica. Taught in 1.5
Max tokens	A ceiling that a caller can set, that says never write more than this. Taught in 20.7
MD5	A real hash function. It gives one big number, written as 32 hexadecimal characters. Taught in 8.4
Megabyte (MB)	1,024 kilobytes. Round it to a million bytes. Taught in 2.2
Memory table (memtable)	The desk: recent changes in memory, used for quick answers. Taught in 6.10
Menu	A small text file that lists the qualities we have, and the small files for each one. Taught in 14.3
Mercator	A crawler that 2 researchers at Compaq built around 1999, to work out how a crawler could grow to the size of the web. Taught in 9.5 A projection named after a mapmaker. Sizes are wrong across the world, and shapes are right up close. Taught in 18.2
Merge	Combining 2 versions into one. Taught in 6.6
Merging	Making two lists that are already in order into one, by repeatedly comparing the two top items. Taught in 11.5
Merkle tree	A tree of fingerprints where each one depends on the ones below it. Named after Ralph Merkle. Taught in 6.9
Message queue	A durable component that helps systems communicate asynchronously. Like a mailbox. Taught in 1.9
Metadata	Information a message carries about itself, like a message ID, who sent it, who it's going to, and a timestamp. Taught in 2.5
Metrics	Measurements over time, like CPU, or daily active users. Taught in 1.9
Microsecond	One millionth of a second. Taught in 2.3

Microservice	A small separate service that does one job, like the search service. `Taught in 4.3`
Middleware	Software that sits between the client and our API servers, and looks at every request before it gets through. `Taught in 4.3`
Millisecond	One thousandth of a second. `Taught in 2.3`
Min in-sync replicas	A floor. If fewer than this many copies are on the list, refuse the write. `Taught in 19.10`
Model	The program that writes the answers. `Taught in 20.1`
Modulo	Dividing and taking the remainder. `Taught in 1.10`
Monthly active users	People who opened the app at least once in the month. `Taught in 2.5`
MQTT	A small protocol designed to go easy on bandwidth and batteries. `Taught in 12.2`
Multi-master replication	The name you might see for skipping numbers. Multi means more than one database that accepts writes. No IDs are actually being copied. `Taught in 7.2`
Multi-region	Deployed in two or more geographic locations around the world. `Taught in 1.6`
Multipart upload	Amazon S3's built-in way to upload in parts, where you only retry the parts that failed. `Taught in 14.4`
MyRocks	MySQL with a different way of storing things on disk underneath. `Taught in 12.4`

N

N	The number of copies of each key. `Taught in 6.3`
Namespace	A named space that things live in. Dropbox's post uses the word for a space like a shared folder. `Taught in 15.3`
Nanosecond	One billionth of a second. `Taught in 2.3`
Network partition	A broken connection between servers. Nothing has crashed, they just can't talk. `Taught in 6.2`
Nginx	Very popular software for web servers and load balancers. `Taught in 4.5`
Node / edge	A dot is a node, and an arrow is an edge. `Taught in 9.4`
Non-functional requirements	How well it has to work, and not what it does. `Taught in 18.1`
Normalization	Cleaning up what someone typed, like making it lowercase and treating ST as street. `Taught in 18.2`
Normalizing	Cleaning up an address before asking if we've seen it, so all of its versions turn into one. `Taught in 9.3`
Northguard	LinkedIn's own new system. `Taught in 19.11`
NoSQL	A non-relational database. Examples: CouchDB, Neo4j, Cassandra, DynamoDB. `Taught in 1.2`
Not before time	The earliest we can go back to a website. `Taught in 9.5`

Notification	A message we send to a person, without them asking for it at that moment. Taught in 10.1
NTP	A way for a computer to ask a trusted clock what time it is. Taught in 7.6
Nudge	A small message that only says something changed, so that the device goes and asks. Taught in 15.6

O

O(1)	Constant. The work doesn't grow with the data. Taught in 13.2
O(log n)	The work grows with how many times you can cut the data in half. Each doubling costs 1 more look. Taught in 13.2
O(n log n)	n times log n. What good sorting costs. Taught in 13.2
O(n squared)	n times n. What comparing everyone with everyone costs. Taught in 13.2
O(n)	Linear. Twice the data, twice the work. Taught in 13.2
OAuth	A standard, and a common way to get a temporary pass instead of handing over your real secret. Taught in 10.2
Object storage	Online storage built for big files, like Amazon S3. Taught in 2.7 A giant folder for files. You give it a file and a name, and later you ask for that name. An object is just a word for a file. It exists because it's way cheaper and easier than keeping billions of files in a database. Taught in 11.2
Offset	How many results to skip. Taught in 3.4
Offset (in a log)	A message's position in the log, like a page number. Taught in 19.2
One-click unsubscribe	Someone can leave a mailing list with a single click, without logging in. Taught in 10.7
Opaque	The store can't see inside the value, like a coat check that never goes through your pockets. Taught in 6.1
Open source	Sharing the code with everyone. Taught in 7.4
OpenStreetMap	A free map of the world that volunteers make and keep up to date. It exists so that nobody has to own the map. Taught in 18.1
Operating system	The software in charge of the whole phone. Taught in 17.8
Operational transformation / CRDTs	2 names for how tools like Google Docs combine small edits. You most probably won't be asked how they work in this question. Taught in 15.7
Opt in	It's off until someone deliberately turns it on. Taught in 17.9
Opt out	Just saying no. Taught in 9.9
Origin	The place where the original file lives, like your web server or Amazon S3. Taught in 1.7
Over-engineering	Building something way more complicated than the problem actually needs. Taught in 3.1

Overloaded error	A message that says you did nothing wrong, and they're just full. `Taught in 20.9`
Overture Maps	A newer project from companies including Amazon, Meta and Microsoft. It combines open map data so that companies can share that work. `Taught in 18.1`
OWASP	A well known non-profit group for web security. `Taught in 11.2`

P

Packet	A small chunk of data that gets sent on its own. `Taught in 2.3`
Page cache	The recently used pieces of files that the operating system keeps in spare memory. `Taught in 19.5`
Page token	What Google Drive's public API gives you to save for next time. It's the same shape as a bookmark. `Taught in 15.6`
PageRank	A score for how important a page is, based on the pages that link to it. `Taught in 9.6`
Pages (memory)	Small pieces of memory that an operating system hands out as they're needed. `Taught in 20.6`
Pagination	Sending the results one page at a time, instead of all of them at once. `Taught in 3.4`
Pagination token	A little marker that tells the next request where to continue from. The same idea as an offset. `Taught in 3.4`
Parsing	Going through a long string of letters and working out the pieces, like reading a recipe and pulling out the ingredients. `Taught in 9.2`
Partition	A stretch of the ring that one server owns. `Taught in 5.4` One piece of a topic, like one volume of a book. `Taught in 19.3`
Partition key	In Cassandra, the first part of the key. It decides which machine, and the second part sorts the rows on that machine. `Taught in 12.4`
Partition tolerance	The system keeps working through a network partition. `Taught in 6.2`
Partitioning (data)	Splitting your data across servers. Each piece is called a partition. `Taught in 6.3`
Path	The address of a file, like book club/week one/notes. `Taught in 15.3`
Payload	The actual content that we're sending. `Taught in 10.2`
Peer to peer	Devices talking straight to each other, with nothing in the middle. A peer just means an equal. `Taught in 17.2`
Per-conversation sequence	Each chat has its own count, so the next message in that chat gets the next number. `Taught in 12.5`
Perplexity	An AI search tool. `Taught in 9.9`
Persistent data store	One that keeps data even when the power goes off. `Taught in 1.6`
Petabyte (PB)	1,024 terabytes, which is about a million gigabytes. `Taught in 2.2`

Pixel	One of the tiny colored dots that a picture is made of. `Taught in 14.2` One of the tiny colored squares that make up a picture on a screen. `Taught in 18.5`
Plus Codes	Google's open source system that turns any spot into a short code, using only math, so a place with no address can still have something you can write down. `Taught in 18.2`
PNG	A common image file. `Taught in 18.5`
Pointer	Where something is in memory. `Taught in 16.7`
Poison message	A bad message that crashes the reader every time, so everyone behind it is stuck. `Taught in 19.11`
Polling	You keep asking on a schedule. `Taught in 12.2`
Polygon	A shape with straight sides. `Taught in 18.5`
Pool	The group of servers behind a load balancer. `Taught in 1.4`
POP	Point of presence. One of a CDN's many locations. `Taught in 18.5`
Port	Which door of a computer you're knocking on. Secure web traffic normally uses door 443. `Taught in 12.2`
PostGIS	An add-on for PostgreSQL, so that a normal database can understand locations and shapes. `Taught in 16.9`
Postmortem	A write-up that says what broke and what they're changing, without blaming anybody. `Taught in 20.10`
Power of two	A number you get by multiplying 2 by itself repeatedly: 2, 4, 8, 16, and 10 times gets you 1,024. `Taught in 2.2`
Pre-signed URL	Amazon's name for a special web address that lets someone upload without having their own credentials. `Taught in 11.2`
Precomputing	Working something out ahead of time. `Taught in 13.6`
Prefix	The beginning of a word. `Taught in 13.1`
Premiere	A video's first public showing. `Taught in 14.6`
Presence	The word for who's online. `Taught in 12.8`
Primary key	A column where every row has a different value, so one number can only mean one row. `Taught in 7.1`
Prime meridian	The line through Greenwich, in London, where longitude is zero. People had to agree on one. `Taught in 16.3`
Priority queue	A line where the most urgent thing goes next, instead of the oldest thing. `Taught in 14.6`
Private IP	An address that only works between machines on the same private network. `Taught in 1.4`
Process	A program that's running on a machine. `Taught in 4.2`

Producer / consumer	Producers (publishers) create messages, and consumers (subscribers) read them and do the work. Taught in 1.9
Projection	A rule that says, this spot on the ball goes to this spot on the paper. Taught in 18.2
Prompt	The whole thing that you send in. Taught in 20.2
Prompt caching	Keeping the model's notes from reading a prompt, and reusing them when a later request has the same beginning. Taught in 20.8
Protocol	A set of rules two sides agree on, like taking turns on a phone call. Taught in 12.2
Proximity service	A service that finds what's close to you. Taught in 16.1
Public IP	An address anyone on the internet can reach. Taught in 1.4
Public key / private key	The padlock and the key. The public key is fine for everybody to have. The private key never leaves your phone. Taught in 12.9
Publish and subscribe	You say once that you want something, and from then on, every new update just shows up, like a newspaper subscription. Taught in 12.8 Publishing means sending something out, and subscribing means asking to receive it. People usually shorten it to pub-sub. Taught in 17.4
Pulsar	An open source system that started at Yahoo. It keeps logs in a separate storage layer, so brokers don't need disks. Taught in 19.11
Push / pull	Push: the broker sends messages as soon as they show up. Pull: the reader asks for them. Taught in 19.7
Push model / pull model	Other names for fan-out on write and fan-out on read. They have nothing to do with push notifications. Taught in 11.4
Push notification	A little message that pops up on your screen from an app, even when the app is closed. Taught in 10.1
Put / get	The 2 operations. Put saves a value under a key, and get gives you the value for a key. Taught in 6.1

Q

QPS	Queries per second. How many operations, like sending a message, happen in each second. Taught in 2.5
Quadkey	A kind of address where each digit says which of the 4 you went into. Taught in 16.9 These same tiles, written as one string of digits. Taught in 18.4
Quadtree	A tree where each square has 4 children, and you only keep cutting where it's crowded. Quad just means four. Taught in 16.4
Query	What someone typed into a search box. Taught in 13.1
Quintillion	A billion billions. Taught in 7.3
Quorum	The rule some clubs have, that a vote only counts if enough members show up. Here, enough servers have to answer. Taught in 6.4

R

R	How many servers have to answer before a read is finished. `Taught in 6.4`
R-tree	A tree of boxes around places. Instead of cutting up the map ahead of time, the boxes follow the data. `Taught in 16.9`
RabbitMQ	A very popular open source message broker. It exists so you don't have to build all that delivering and retrying yourself. `Taught in 19.1`
Race condition	The result depends on the timing of two things happening at the same moment. `Taught in 4.7`
Rack	A tall cabinet of servers in a data center. If its power fails, everything in it goes together. `Taught in 19.9`
Radio	The part of the phone that talks to the network. `Taught in 17.2`
Radius	How far you're searching from the middle. `Taught in 16.1`
Raft	A well known set of rules for a group of machines to agree on one leader and one history. `Taught in 19.9`
RAID	Redundant array of independent disks. A way to make several disks act like one, usually so one can die without losing anything. `Taught in 19.5`
RAM	Random access memory. The computer's short-term memory. `Taught in 1.3`
Ranges	Giving each server a block of numbers to count through on its own, instead of asking for one number at a time. `Taught in 8.6`
Ransomware	A kind of attack that scrambles your files and asks for money. `Taught in 15.9`
Raster image	An image stored as a grid of pixels. Raster just means a grid. `Taught in 18.5`
Rate limiter	Decides how many requests a client is allowed to send over a certain period of time. Extra requests get blocked. `Taught in 4.1`
Read operations	Anything that only looks at data. `Taught in 1.5`
Read-heavy	Way more reads than writes. `Taught in 16.1`
Rebalancing	Working out who reads what. `Taught in 19.8`
Recursion	When a program does the same thing to each smaller piece, again and again. `Taught in 16.7`
Red flag	Something you do that makes the interviewer worried about hiring you. `Taught in 3.1`
Redirect	Like a note on a shop door saying we've moved. The server doesn't fetch the page, it tells the browser where to go. `Taught in 8.2`
Redis	An in-memory database that's very fast. `Taught in 1.8` Remote Dictionary Server. A key-value store in memory that you can run yourself, or get as a managed service from cloud providers such as AWS and Microsoft. `Taught in 4.3`
Redundancy	Having multiple copies of a resource. Having the spare tire. `Taught in 1.3`

Refill rate	How fast tokens come back. `Taught in 4.5`
Relational database (RDBMS)	Stores data in tables and rows, like a spreadsheet. Also called a SQL database. `Taught in 1.2`
Relative link	A link that isn't a full address, like saying the room down the hall. It only makes sense if you know which building you're in. `Taught in 9.2`
Release	Putting a new version of our code on the servers. `Taught in 16.8`
Remainder	What's left over after dividing. `Taught in 5.1`
Replay	Moving your bookmark back and reading those messages again. `Taught in 19.2`
Replica	A copy of the data. The copy that takes writes is the leader, and the others are followers. `Taught in 19.9`
Request body	The part of a request that carries what you're sending. `Taught in 11.2`
Requirement	Something our design has to do. `Taught in 4.2`
Resolution	How many dots a picture has. 1080p is 1,080 rows, each 1,920 dots wide. 4K has about 4 times the dots. `Taught in 14.2`
REST	A popular style of building web APIs, where each thing has its own address and you use a few standard words. `Taught in 8.2`
Resumable upload	After an interruption, the client asks how many bytes have arrived, and continues from the next one. `Taught in 14.4`
Retention	How many of your users keep coming back. `Taught in 1.9`
Retrieval (RAG)	Fetching what's relevant before you ask. RAG stands for retrieval augmented generation. `Taught in 20.5`
Retry storm	Everything that failed tries again at once, so a struggling service gets way more traffic than usual. `Taught in 10.5`
Retry topic	A separate log for things to try again later, so that the main line keeps moving. `Taught in 19.11`
Retry-After	A standard, optional header that tells the client how long to wait before trying again. `Taught in 4.6`
Reverse chronological	The formal name for newest first. `Taught in 11.1`
Reverse geocoding	Going from a dot to words, like showing you what street you're on. `Taught in 18.2`
Road segment	A piece of a road, usually from one junction to the next. `Taught in 18.9`
robots.txt	A small file where a website says what it wants crawlers to do. It's voluntary. `Taught in 9.5`
RocketMQ	An open source system that came out of Alibaba's online stores, and that also keeps messages in logs. `Taught in 19.11`
Rolling checksum	What rsync uses. A checksum is just a short number used to check data. `Taught in 15.4`
Rolling release	Restarting a few servers at a time. `Taught in 16.8`

Root	The one number at the top of the tree. `Taught in 6.9`
Round	One time through. `Taught in 6.7`
Round robin	Taking turns: the first message goes in volume 0, the next in volume 1, and so on. `Taught in 19.3`
Round trip	A trip across the network there and back. It has a cost that doesn't care how small your message is. `Taught in 19.6`
Routing tiles	Squares that the road network is cut into, in 3 sets from every street to only highways. `Taught in 18.7`
rsync	A free tool that exists to update a file on another machine by only sending what's different. `Taught in 15.4`

S

S2	Google's library for working with data on the real, round planet. `Taught in 16.9`
Salt	The extra bit you add to the input before hashing again, like the number 1. `Taught in 8.4`
Sampling	Only writing down some of the data, picked at random, like 1 search in 10. `Taught in 13.7`
Scanning	Trying code after code. `Taught in 8.6`
Scheduler	A small program that watches the graphs, and puts a task in line when it's free to start. `Taught in 14.6`
Schema	The tables and columns. Each shard has the same schema, with different data. `Taught in 1.10`
Scope	What's included in the thing we're building, and what's not. `Taught in 3.2`
ScyllaDB	A database built to work like Cassandra. `Taught in 12.4`
Search index	The collection of pages a search engine can search through. `Taught in 9.1`
Search service	A separate part of our system whose only job is finding books. `Taught in 3.3`
Secret key	Like a password that one program uses to prove who it is to another program. `Taught in 4.1`
Seed URLs	A few starting addresses, called seeds because everything else grows from them. `Taught in 9.2`
Segment	One of the smaller files that a volume is cut into. `Taught in 19.5`
Selector	Takes links out of the priority lines, picking from high most often. `Taught in 9.6`
Sentinel	A Redis tool that watches its servers and can promote a copy automatically. `Taught in 17.7`
Sequence number	The count section of the ID. It goes back to 0 every new millisecond. `Taught in 7.4`

Sequential / random	Reading or writing in one straight line, or jumping around. Taught in 19.5
Serialization	Converting something, like an object in your code, into a format that can be stored or sent, like JSON. Deserialization is the opposite. Taught in 1.2
Server	A computer, or a program on one, that provides services, data or resources to other computers. Taught in 1.1
Server-sent events	Only the server talks, over a normal HTTP connection. Taught in 12.2
Server-side rendering	The website's own server builds the full page before sending it. Taught in 9.8
Service credits	Money off your future bills. Taught in 2.4
Service discovery	A way of finding out which servers exist, and which one to use. Taught in 12.3
Service provider	The company that runs and offers a service, like AWS, Google Cloud or Microsoft Azure. Taught in 2.4
Session registry	Our front desk: one simple lookup of which chat server each person is on. Taught in 12.3
Set	A hash table for the question, have I seen this? We only care whether the hook has been used. Taught in 9.3
SHA-256	A long and strong hash, which Dropbox's posts say it uses. Taught in 15.9
Shard exhaustion	When some shards fill up much faster than others. Taught in 1.10
Sharding	Horizontal scaling for databases. Splitting a large database into smaller parts called shards. Taught in 1.10
Sharding key	One or more columns that determine how data is distributed. Also called a partition key. Taught in 1.10
Share groups	A Kafka feature where members of a group can share one partition, message by message, like a mailbox. Taught in 19.11
Shared access signature	Microsoft's name for its version of a pre-signed URL. Taught in 14.4
Shifting	Sliding bits to the left to leave room, like adding 3 zeros after 0930. Each place is the same as doubling the number. Taught in 7.5
Shortest path	In this chapter, the smallest total time. Taught in 18.6
Signal Protocol	A published design for this kind of encryption. WhatsApp's paper says it's built on it. Taught in 12.9
Signed cookie	A cookie is a small note your browser keeps, and sends with every request to that site. A signed one can cover all of a video's pieces. Taught in 14.8
Signed link	A special address that can't be changed without us noticing, and that expires soon. Taught in 14.8
Silent push	A kind of push with nothing to show, which wakes your app up for a moment. Taught in 17.8
SimHash	A way of making a fingerprint where similar pages get almost the same number, so you can spot near copies. Taught in 9.3

Single point of failure	One part of a system that takes the whole system down when it fails. Taught in 1.3
SLA	Service level agreement. A formal contract that says how much uptime a customer should expect, and what happens if the provider misses that. Taught in 2.4
Sliding window counter	Keeps counts for the current and previous minutes, and estimates how many requests are in the last 60 seconds. Taught in 4.4
Sliding window log	Keeps a log of timestamps for each customer, and looks back at exactly the last 60 seconds on every request. Taught in 4.4
Sloppy quorum	Counts the first healthy servers, even if they aren't the usual ones. Taught in 6.8
SMS	Short message service. A text message, sent over the phone network to a phone number. Taught in 10.1
Snapshot	A picture of everything at one moment, like a saved copy of a trie. Taught in 13.7
Snowflake	An ID layout built at Twitter: sections for time, place and a count, in 64 bits. Taught in 7.4
Soft limit	A limit that can be passed for a short time. Taught in 4.8
Soft real time	We try to be quick, and nothing is broken if we're a few seconds behind. Taught in 10.1
Solid state drive	A drive with no moving arm, where the gap between sequential and random is a lot smaller. Taught in 19.5
Sorted set	Like a set, so there are no duplicates, except each item has a number called a score, and everything stays in order of that number. Taught in 11.4
Spam	Pages stuffed with popular words to fool search engines. Taught in 9.8
SPF, DKIM and DMARC	Three checks that prove an email really came from your domain, so nobody can pretend to be you. Taught in 10.7
Spider trap	A path of pages a crawler can follow forever. A crawler is sometimes called a spider, and some people build these on purpose. Taught in 9.8
SQL	The language you use to talk to a relational database. Taught in 1.2
SSTable	Sorted string table. A box on disk that's sorted and never changed. Taught in 6.10
Stack	A pile of plates. The last one on is the first one off. Taught in 9.4
Staging	A practice copy of a system. Taught in 20.10
Stale data	Data that's out of date. Taught in 1.6
Standard deviation	A measure of how spread out numbers are around their average. Taught in 5.4
Standby	A copy that's kept up to date and ready to take over. Taught in 17.7
State / session data	Information a server remembers about a user, like that you're logged in. Taught in 1.8
Stateful server	Remembers client data from one request to the next. Taught in 1.8

Stateless server	Keeps no state information. `Taught in 1.8`
Static content	Files that are the same for everyone, like images, videos, CSS and JavaScript. `Taught in 1.7`
Status code	The number every answer from a web server starts with. The first digit tells you what kind of answer it is. `Taught in 8.2`
Sticky sessions	The load balancer sends every request from one client to the same server. `Taught in 1.8`
Stream	Data that keeps arriving and doesn't really have an end. `Taught in 13.10`
Streaming	Sending the answer in pieces as it's made. `Taught in 20.4`
Strict quorum	Only counts the usual servers, like A, B and C. `Taught in 6.8`
Strong consistency	Once a change is finished, every read, from anyone, gets the new data. `Taught in 6.5`
Subnet	A range of addresses that defines which network machines belong to. `Taught in 1.4`
Surge pricing	The higher price when there are too many riders and not enough cars. `Taught in 16.9`
Sync	Keeping copies in different places the same. `Taught in 15.1`
Synchronous	The sender waits until the receiver is done. Like calling a friend. `Taught in 1.9`

T

TCP	Gets data across the internet reliably, with all the pieces, in the right order. `Taught in 12.2`
Template	A message with blanks in it, filled in at the last moment. `Taught in 10.7`
Temporal	An open source platform that exists so that a long-running job can keep its progress when something fails. `Taught in 14.6`
Terabyte (TB)	1,024 gigabytes. Round it to a trillion bytes. `Taught in 2.2`
Theorem	A rule that somebody has proven. `Taught in 6.2`
Third party	Someone outside of our company doing part of our job. `Taught in 10.2`
Thread	One worker inside a program, doing one thing at a time. `Taught in 9.5`
Throttling	Limiting someone's requests. If a user is throttled, the rate limiter stepped in. `Taught in 4.1`
Throughput	How much gets through in a second. `Taught in 19.1`
Thumbnail	A tiny, blurry version of a picture. `Taught in 12.7`
Thundering herd	Adele's post expires from the cache, thousands of people ask for it, and they all go to the database at once. A common fix is to let one request go get it, while everyone else waits a moment. You probably won't get asked this. `Taught in 11.6`

Ticket server	A small database with a counter, whose only job is to hand out numbers. Like the ticket machine at a deli counter. `Taught in 7.2`
Tier	One of the levels that companies set their limits at. `Taught in 20.7`
Tiered storage	Where finished segments can go to something like S3, while the newest stay on local disks. `Taught in 19.5`
Time to first token	How long until the first token shows up. `Taught in 20.3`
Time wheel	A common way to track a lot of timers. One bucket per second, and a hand moves around and empties each one. `Taught in 19.11`
Timeout	How long we're willing to wait before giving up. `Taught in 9.7`
Timestamp	A record of the exact time that something happened. `Taught in 4.4`
TLS	The technology underneath HTTPS. SSL is the old name, and its last version was officially ruled out in 2015, so please say TLS. `Taught in 15.8`
Token	A small piece of text. Sometimes a whole word, sometimes part of a word, and sometimes only a comma. `Taught in 20.2`
Token bucket	Each request pays with a token from a bucket that refills at a steady speed. Empty means rejected. `Taught in 4.5`
Tombstone	A small marker that says this was deleted. `Taught in 6.11`
Top K	Give me the K most popular things. K is just a number, and for us it's 5. `Taught in 13.1`
Topic	A name for one stream of messages, like one book. `Taught in 19.2`
TPU	Tensor processing unit. Google's custom chip for machine learning. `Taught in 20.2`
Tracking parameters	Small notes a link carries after the question mark, so a company knows where you came from. `Taught in 8.1`
Tracking pixel	A tiny invisible image in an email. When your mail app loads it, the sender finds out. `Taught in 10.8`
Tradeoff	What you give up in order to get something. `Taught in 3.1`
Trading space for time	We spend memory, which we can buy, and we get back time. `Taught in 13.6`
Training	Teaching the model by practicing on a huge amount of text. It's a giant project that happens before you ever meet the model. `Taught in 20.2`
Transactions (Kafka)	A program reads from some topics, writes to others and moves its bookmark, all as one step that either all happens or doesn't. `Taught in 19.10`
Transcoding	Turning one encoded video into a different one: a different codec, resolution or bitrate. `Taught in 14.2`
Transfer Acceleration	An Amazon feature where data goes in at a nearby CDN location, then travels over a better path to storage. Kind of a CDN in reverse. `Taught in 14.4`
Traveling salesman problem	Finding which order a vehicle with a lot of stops should go in. `Taught in 18.12`

Tree	A shape of fingerprints that branches out, even though it's upside down. Taught in 6.9 Things hang off each other, with the middle at the top, what comes before it to the left and what comes after it to the right, so each step cuts what's left in half. Taught in 13.3
Triage	The desk at the front of an emergency room, where a nurse decides how urgent you are. Taught in 9.6
Trie	A tree made out of letters, where the path from the root spells out what you've typed. Spelled T R I E. Taught in 13.5
Trilateration	Finding someone from their distance to 3 different spots. Taught in 16.11
TTL	Time to live. How long a file should stay cached. Taught in 1.7
Tunable consistency	Being able to tune how in sync the copies are. Taught in 6.1
Turn	One question and its answer. Taught in 20.5
tus	An open protocol that exists so that any app can have resumable uploads over normal HTTP, without building its own. Taught in 14.4
Twilio	A company that businesses pay to send texts for them. Taught in 10.1
Two-dimensional	You need 2 numbers to say where something is, like a row and a column to find a seat. Taught in 16.3
Two-way sharing	Friends who agree to see each other. In one-way sharing, I can see you and you can't see me. Taught in 17.11

U

Unicode	The standard that gives a number to the characters of all the world's writing systems, with room for over a million. Taught in 13.10
Uniqueness rule	Something you tell the database about a column: no 2 rows can ever have the same value here. Taught in 8.4
Unix time	Computers usually count time from January 1, 1970. Taught in 7.5
Unstructured data	Data that doesn't fit neatly into tables. Taught in 1.2
Uptime	The time that the system is up and working. Taught in 2.4
URL	Uniform resource locator. The full address of a page on the internet. Taught in 8.1
URL frontier	The crawler's to-do list. It holds every link we want to visit and haven't visited yet. Taught in 9.2
URL Seen	The box that asks: have we already got this address? Taught in 9.2
Use case	What something is actually going to be used for. Taught in 4.2
UTC	One clock that the whole world agrees on, so that time zones don't confuse anything. Taught in 7.5

UTF-8	The usual way of storing Unicode, taking between 1 and 4 bytes for each character. Taught in 13.10
UUID	Universally unique identifier. A 128-bit number so big and so random that you can make one whenever you want. Taught in 7.3

V

Vector clock	A small list of servers and write counts that travels with each version. Vector means a list of numbers, and it tells you the order things happened in without looking at the time. Taught in 6.6
Vector tile	A tile sent as a list of what's in it. Vector means it's made of lines and shapes, not pixels. Taught in 18.5
Versioning	Adding a version number to a file's address, so the CDN treats it as new. Taught in 1.7
Viewport	The part of the map that's on your screen right now. Taught in 18.4
Virtual node	One more place on the ring for a real machine. It isn't a real machine, and it isn't a copy of any data. Taught in 5.4
Visibility timeout	How Amazon's ready-made queue service, SQS, hides a message from everyone else for a while. It shows up again if the worker didn't finish. Taught in 14.6
vLLM	An open source program for running these models, which started at UC Berkeley, so you don't have to build all of this yourself. Taught in 20.6
Volatile	Lost when a server restarts, as data in memory is. Taught in 1.6
VP9	A newer codec from Google, free to use. Taught in 14.2

W

W	How many servers have to say got it before a write is finished. Taught in 6.4
Watermark	A mark inside the video that says whose it is: a logo you see, or a hidden pattern that's different for each customer. Taught in 14.8
Waze	A navigation app that's built around drivers reporting what they see. Taught in 18.8
Weak consistency	You might see new data or old data, and the system doesn't tell you when that changes. Taught in 6.5
Web crawler	A program that goes and gets pages from around the web: download, pull out the links, add them to the list, and repeat. Taught in 9.1
Web Mercator	The version nearly every web map uses. It's Mercator with a small simplification that treats the Earth as a perfect ball. Taught in 18.2
Web tier / data tier	The layer handling web and mobile traffic, and the layer holding the database. Taught in 1.1

WebRTC	Web real-time communication. It exists so that browsers and apps can send live sound and video directly to each other, often without a server in the middle. `Taught in 14.9`
WebSocket	A protocol built so that either side can talk whenever they want, over a connection that stays open. `Taught in 12.2`
Weight	The number on an arrow. Here, how many minutes we expect that road to take. `Taught in 18.6`
Wide-column store	A database where one key can hold many rows, kept in sorted order. `Taught in 12.4`
Widevine / FairPlay / PlayReady	The DRM systems from Google, Apple and Microsoft. To reach every device, you often need more than one. `Taught in 14.8`
Window	The stretch of time that we're counting in. `Taught in 4.4`
Worker	A program that takes jobs from a queue and does them. `Taught in 10.4`
Write operations	Anything that changes data, like inserts, updates and deletes. `Taught in 1.5`

Y

YAML	A text format that's an easy way to write settings so that people can read them. `Taught in 4.6`

Z

Zanzibar	A system Google published in 2019, so that its products can share one way of storing and checking these lists. You probably won't get asked this. `Taught in 15.8`
Zero copy	Telling the operating system to send bytes straight from the page cache to the network, so our program doesn't touch them. It can't be used when the connection's encrypted with SSL. `Taught in 19.6`
Zipf	A common pattern where number 2 is half as popular as number 1, number 3 gets a third, and so on. `Taught in 14.7`
ZooKeeper	A small and very reliable place where machines can agree on things. You don't need to know how it works. `Taught in 7.6`
Zoom level	One copy of the whole map, at one distance from far away to very close. `Taught in 18.4`